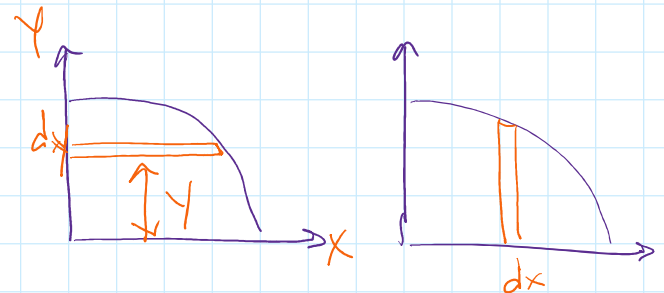


Front Page

- * Study 1st 3 Exams C.G.
 - * Physical significance of a Centroid
 - * How are centroid, center of mass & center of gravity related?
 - * Physical significance of mass moment of inertia?
 - * Radius of gyration (what does it represent)
 - * How does mass moment of inertia change as the mass moves away from, or closer to, an axis.
1. Ch.4 / Ch.5 Problem. Reaction @ supports. Distributed loading (integrate). Pinned, roller, fixed support.
 2. Ch.8 - Friction Problem. Impending motion may be present. $F_s = \mu N$
Rigid body. May include spring. $F = k(l - l_0)$
Reaction @ supports. amount of stretch
 3. Ch.9 / Ch.10 - Centroids & moments of inertia
Two-dimensional (integration)

horizontal rectangle $\rightarrow$ $I_x = \int y^2 dA$

vertical rectangle $\rightarrow$ $I_y = \int x^2 dA$



$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA}$$

$$\bar{y} = \frac{\int \tilde{y} dA}{\int dA}$$

* look @ 9.29 & 10.22 for practice.